



Roll Number

NARULA INSTITUTE OF TECHNOLOGY
AN AUTONOMOUS INSTITUTE UNDER MAKAUT
2025
2024-25 EVEN SEMESTER
PH(CS)201 - ENGINEERING PHYSICS

TIME ALLOTTED: 3Hours**Maximum Marks : 70****Instructions to the candidate:***Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. A key characteristic of laser light is its (1) C01 BL1 PO1
(a) Polychromatic nature (b) Incoherence
(c) High directionality (d) Low intensity
2. How many valence electrons do trivalent impurities have? (1) C03 BL1 PO1
(a) 2 (b) 3
(c) 4 (d) 5
3. The example of wave-particle duality is (1) C05 BL2 PO1
(a) A moving Electron (b) a rest rock
(c) a cricket bat (d) None of the above.
4. What is the full form of CNT? (a) Carbon Nanotube (b) Car Nanotube (c) Carbon Non-tube (d) Carbon New tube (1) C02 BL1 PO1
5. What type of device is a pen drive? (1) C04 BL1 PO1
(a) Semiconductor storage device (b) Optical storage device
(c) Magnetic storage device (d) None of the above
6. The velocity with which energy is transported through a medium is (1) C01 BL2 PO2
(a) Group velocity (b) Phase velocity

- (c) Complex velocity (d) Imaginary velocity
7. The devices which converts the electrical energy into light is called (1) C03 BL1 PO1
- (a) LED (b) Non-emitters
(c) Plasma panels (d) Emitters
8. Strongest material of the world is (1) C01 BL1 PO2
- (a) Carbon steel (b) Graphene
(c) Steel (d) ALNICO
9. The Light emitting diode (LED) is made of (1) C01 BL2 PO2
- (a) Si (b) Ge
(c) GaN (d) Tungsten
10. The number of Bravais lattice in three dimensions are (1) C01 BL1 PO1
- (a) 7 (b) 9
(c) 14 (d) 19
11. Miller indices (hkl) represent (1) C01 BL2 PO2
- (a) A set of parallel plane (b) A particular plane
(c) A set of arbitrary oriented planes (d) None of these
12. In optical fibre the transmission medium is : (1) C01 BL1 PO1
- (a) electric waves (b) radio waves
(c) light (d) none

Part B**(Answer any three of the following) 3x5=15**

13. What is Hologram? Explain the construction and reconstruction of the hologram with proper diagram. (5) C01 BL2 PO1
14. What is unit cell? Find the Miller indices of a plane with intercepts 3a, 2b and c. (5) C03 BL2 PO1
15. Explain the phenomenon of Compton effect. Calculate the Compton wave length of electron (5) C05 BL3 PO2
16. How do the properties of diamond and graphite differ significantly? Mention one important application of carbon nanotubes. (5) C02 BL1 PO1

17. Write down the advantage and disadvantages of Magnetic storage devices? (5) C04 BL1 PO1

Part C**(Answer any three of the following) 3x15=45**

18. (a) Define the coordination number. Write the coordination number for atoms in an SC and BCC lattice. (3) C03 BL1 PO1
- (b) Calculate Atomic Packing Factor (APF) for a simple cubic structure. How it is related to void space? (5) C03 BL1 PO1
- (c) State Bragg's law. Explain the significance of Bragg's law in X-ray crystallography. (4) C03 BL1 PO1
- (d) A crystal with an interplanar spacing of $d=2.5 \text{ \AA}$ produces a diffraction peak at an angle of $\theta=30^\circ$. Calculate the wavelength of the X-rays used. (3) C03 BL4 PO1
19. (a) Find the Compton wavelength of an electron. (3) C05 BL1 PO1
- (b) Write the time-independent Schrodinger wave equation. (3) C05 BL1 PO1
- (c) What is a wave function? Write the significance of the wave function. (3) C05 BL1 PO1
- (d) Write the experimental outcome of Davisson-Germer's experiment. (3) C05 BL1 PO1
- (e) Calculate the value of the de Broglie wavelength of an electron moving under the potential difference of 300 volts. (3) C05 BL2 PO1
20. (a) Describe Quantum confinement. (3) C02 BL3 PO1
- (b) Distinguish between Quantum well, Quantum wire and Quantum dots. (6) C02 BL3 PO1
- (c) Explain the operation and application of CRT, LCD and OLED. (6) C04 BL4 PO1
21. (a) What is LASER? Write the basic properties of LASER. (3) C01 BL1 PO1
- (b) Derive the relation between Einstein's A and B Coefficients. (6) C01 BL1 PO1
- (c) What is Population Inversion? (3) C01 BL1 PO1
- (d) What is a Meta-stable state? (2) C01 BL1 PO1
- (e) What is the value of LASER wavelength in Ruby LASER? (1) C01 BL1 PO1
22. (a) What is the difference between Hard disc and Pendrive? (2) C03 BL2 PO2

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| (b) Write down the working principle of CRO? | (6) | C04 | BL2 | PO2 |
| (c) Write down the applications of CRO? | (4) | C03 | BL2 | PO3 |
| (d) Give a short note on LCD | (3) | C02 | BL2 | PO2 |